Yanbang Wang

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Google Scholar

SUMMARY

Machine-learning PhD researcher specializing in **Recommendation Systems**, **Data Agents**, and **Information Retrieval**. At Google, led and first-authored **Gemini-SQL2**, Google's state-of-the-art foundation model for data agents, ranking **#1 on the BIRD Bench**, the de-facto standard benchmark for data agents. Working full-stack across **agentic harness, post-training, and evaluation**. 7 First-authored publications at NeurIPS (2), ICML (1), ICLR (2), and WebConf (2) on **large-scale recommendation system**, **LLM reasoning, retrieval, and verification over structure-rich data**, and **deep learning on graphs**. Internships at Google, Meta, and Microsoft.

EDUCATION

2025 - Present — Visiting Ph.D. Student
Stanford University, Stanford, CA

2021 - Present — Ph.D. in Computer Science
Cornell University, Ithaca, NY
Advisor: Jon Kleinberg

2019 - 2021 — M.S. in Computer Science
Stanford University, Stanford, CA
Advisor: Jure Leskovec

2015 - 2019 — B.S. in Computer Science, Mathematics
Hong Kong University of Science and Technology, Hong Kong, Hong Kong

PROFESSIONAL EXPERIENCE

2025 – 2026 — Google Research, Research Intern

- Led and first-authored Gemini-SQL2, Google's state-of-the-art text-to-SQL coding LLM – currently #1 on the industry-standard BIRD Bench (single-model track). (Google Announcement, Post 2, Post 3, Post 4)
- Post-trained Gemini 3.1 Pro: SFT, RLVR, Param-efficient adaptor tuning, data synthesis.
- Worked full-stack on the agentic harness, including context retrieval, code analysis, prompt engineering, execution feedback, self-refinement loop, test-time scaling, validation, and eval pipeline.

2025 — Meta AI, Research Intern

- Redesigned the Transformer backbone for re-ranking for Facebook videos personalization and recommendation.
- Successfully shipped – improving personalized video experience of all Facebook users (∼3.1 billion).

2024 — Meta AI, Research Intern

- Researched and developed provably optimal negative-sampling theories for contrastive learning in recommendation systems.
- Published as first-author an ICML paper [C.1].

2023 — Microsoft Research, Research Intern

- Researched and developed temporal graph-learning models for retrieving and detecting anomalies from network communication log, for Microsoft Defender and Microsoft 365.
- Published as first-author a WebConf paper [C.5] and a patent [PA.1].

PUBLICATIONS & PATENTS

(Citations: 1546, H-Index: 12)

PREPRINTS & MANUSCRIPTS

[P.1] **Yanbang Wang**, Qitian Wu, Sami Abu-el-Haija, Mohammadreza Pourreza, Michael Galkin, Hadi Hemmati, Hailong Li, Yeounoh Chung, Fatma Ozcan, Bryan Perozzi, Vahab Mirrokni. Gemini-SQL2: Model, Harness, and System Design. *Under Review*.

[P.2] **Yanbang Wang**, Sami Abu-el-Haija, Mohammadreza Pourreza, Michael Galkin, Hadi Hemmati, Yeounoh Chung, Fatma Ozcan, Bryan Perozzi. Graph-Language Models as Text-to-SQL Verifier. *Under Review*.

[P.3] Sami Abu-el-Haija, Fatma Ozcan, **Yanbang Wang**, Brandon Mayer, Bryan Perozzi. Bidirectional Graph–Code Reasoning: Generating and Validating Queries through Structural Representations. *Under Review*.

PEER-REVIEWED CONFERENCE PUBLICATIONS

[C.1] **Yanbang Wang**, Jon Kleinberg, Yanhong Wu. Negative Sampling From the Ground Up: a Redesign for Recommendation. Proceedings of the 43rd International Conference on Machine Learning **(ICML)**, 2026.

[C.2] **Yanbang Wang**, Hejie Cui, Jon Kleinberg. Microstructures and Accuracy of Graph Recall by Large Language Models. Advances in Neural Information Processing Systems **(NeurIPS)** 38, 2025.

[C.3] **Yanbang Wang**, Jon Kleinberg. From Graphs to Hypergraphs: Hypergraph Projection and its Reconstruction. Proceedings of the 12th International Conference on Learning Representations **(ICLR)**, 2024.

[C.4] **Yanbang Wang**, Jon Kleinberg. On the Relationship Between Relevance and Conflict in Online Social Link Recommendations. Advances in Neural Information Processing Systems **(NeurIPS)** 37, 2023.

[C.5] **Yanbang Wang**, Karl Hallgren, Jonathan Larson. A Graph-based Framework for Reducing False Positives in Authentication Alerts in Security Systems. Companion Proceedings of the International World Wide Web Conference **(WebConf)**, 2023 .

[C.6] Haoteng Yin, Muhan Zhang, **Yanbang Wang**, Jianguo Wang, Pan Li. Algorithm and system co-design for efficient subgraph-based graph representation learning. Proceedings of the VLDB Endowment **(VLDB)**, 2022.

[C.7] **Yanbang Wang**, Yen-Yu Chang, Yunyu Liu, Jure Leskovec, Pan Li. Inductive Representation Learning in Temporal Networks via Causal Anonymous Walks. Proceedings of the 9th International Conference on Learning Representations **(ICLR)**, 2021.

[C.8] **Yanbang Wang***, Pan Li*, Chongyang Bai, Jure Leskovec. TEDIC: Neural Modeling of Behavioral Patterns in Dynamic Social Interaction Networks. Proceedings of the International World Wide Web Conference **(WebConf)**, 2021 .

[C.9] Pan Li, **Yanbang Wang**, Hongwei Wang, Jure Leskovec. Distance Encoding: Design Provably More Powerful Neural Networks for Graph Representation Learning. Advances in Neural Information Processing Systems **(NeurIPS)** 34, 2020.

PEER-REVIEWED JOURNAL PUBLICATIONS

[J.1] Haipeng Zeng, Xinhuan Shu, **Yanbang Wang**, Yong Wang, Liguo Zhang, Ting-Chuen Pong, Huamin Qu. EmotionCues: Emotion-Oriented Visual Summarization of Classroom Videos. IEEE Transactions on Visualization and Computer Graphics **(TVCG)**, 2020.

WORKSHOPS & EXTENDED ABSTRACTS

[W.1] **Yanbang Wang**, Hejie Cui, Jon Kleinberg. Network Recall by Large Language Models. International Conference on Computational Social Science **(IC2S2)**, 2024. *Oral.*

[W.2] **Yanbang Wang**, Jon Kleinberg. On the Relationship Between Relevance and Conflict in Online Social Link Recommendations. International Conference on Computational Social Science **(IC2S2)**, 2024.

[W.3] Haoteng Yin, **Yanbang Wang**, Pan Li. Revisiting Graph Neural Networks and Distance Encoding in a Practical View. Deep Learning on Graphs: Method and Applications **(AAAI-DLG)**, 2021.

[W.4] **Yanbang Wang**, Pan Li, Chongyang Bai, VS Subrahmanian, Jure Leskovec. Generic Representation Learning for Dynamic Social Interaction. 19th International Workshop on Mining and Learning with Graphs **(KDD-MLG)**, 2020.

[W.5] **Yanbang Wang**, Bijia Chen, Cameron Campbell, A Network-based Method for Estimating Potential for Career Advancement from Incomplete Data. Social Science and History Association Annual Meeting **(SSHA)**, 2020.

PATENTS

[PA.1] Karl Hallgren, **Yanbang Wang**, Jonathan Larson. Network Authentication Evaluation. United States Patent and Trademark Office, Patent Application No. US20250337760A1, 30 Oct. 2025.

TECHNICAL SKILLS

Languages	Python, C++
ML / LLM	PyTorch, JAX, Numpy, PEFT, SFT, RLVR
System / Tools	Google-internal tools & workflows all suite (XManager, Fig/Piper, CNS, Critique, Cider, Jetski/Gemini Coder etc.), Google Cloud Platforms, Vertex AI Model Endpoint Deployment

ACADEMIC SERVICE

GENERAL CHAIR

2025	Learning on Graphs Conference (LoG 2025)
2026	Learning on Graphs Conference (LoG 2026)

PROGRAM COMMITTEES / REVIEWER

(ML: Machine Learning; DM: Data Mining; CSS: Computational Social Science)

ML	NeurIPS 2021- 2026, ICLR 2022 - 2025, ICML 2023 - 2026, LoG 2022 - 2026
DM	KDD 2023, SDM 2024, Journal of Big Data
CSS	IC2S2 2024

SELECTED HONORS AND AWARDS

2026 Thinking Machine - Tinker Research Grant
2025 Vessel AI Research Grant
2024 Microsoft Accelerating Foundation Models Research Grant
2024 ICLR Student Travel Award
2023 NeurIPS Student Travel Award
2021 Stanford Graduate with Distinction in Research Honor
2019 HKUST Academic Achievement Medal (highest undergraduate honor, top 1%)
2019 HKUST Outstanding Graduate

TALKS AND PRESENTATIONS

INVITED TALKS

2026 Google Research, GenAI
2025 Cornell Theory Seminar
2025 Meta AI
2023 Microsoft Research
2021 Tsinghua University, AI-Time Forum
2021 Tianjin University, College of Intelligence and Computing
2020 UIUC, Data Mining Group (DMG)
2018 MIT CSAIL, Any-scale Learning For All Group (ALFA)